Diversity of Carbapenem Resistance Mechanisms in Clinical Gram-Negative Bacteria in Pakistan

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Shahid ahmad Abbasi
Fauji Foundation Hospital and Medical College

Bushra Jamil
BJ Micro Lab (SMC Private) Limited

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MECHANISMS

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Diversity of Carbapenem Resistance Mechanisms in Clinical Gram-Negative Bacteria in Pakistan

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Linda Hadjadj,[1] Muhammad Ali Syed,[2] Shahid Ahmad Abbasi,[3] Jean-Marc Rolain,[1] and Bushra Jamil[4]

Antibiotic resistance is a health challenge worldwide. Carbapenem resistance in Gram-negative bacteria is a major problem since treatment options are very limited. Tigecycline and colistin are drugs of choice in this case, but resistance to these drugs is also high. The aim of this study was to describe the diversity of resistance mechanisms in carbapenem-resistant clinical Gram-negative bacteria from Pakistan. Carbapenem-hydrolyzing enzyme-encoding genes were detected using PCR and DNA sequencing and clonal types determined by multilocus sequence typing (MLST). Forty-four carbapenem-resistant isolates were collected from the microbiology laboratory of Fauji Foundation Hospital and Al-Syed Hospital, Rawalpindi, Pakistan, including *Klebsiella spp.*, *Escherichia coli*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, *Enterobacter cloacae*, and *Achromobacter xylosoxidans*. $bla_{\text{NDM-1}}$, $bla_{\text{NDM-4}}$, $bla_{\text{NDM-5}}$, $bla_{\text{NDM-7}}$, $bla_{\text{OXA-48}}$, and $bla_{\text{OXA-181}}$ were detected in Enterobacteriaceae; $bla_{\text{OXA-23}}$, $bla_{\text{OXA-72}}$, and $bla_{\text{NDM-1}}$ in *A. baumannii*, and $bla_{\text{VIM-6}}$ and $bla_{\text{VIM-11}}$ in *P. aeruginosa*. MLST analysis revealed several predominant clonal types: ST167 in *E. coli*, ST147 in *Klebsiella pneumoniae*, ST2 in *Acinetobacter*, and ST664 in *P. aeruginosa*. In *Acinetobacter*, a new clonal type was observed for the first time. To the best of our knowledge, this is the first study describing the clonality and resistance mechanisms of carbapenem-resistant
AU4 ▶ Gram-negative bacteria in Pakistan.

Keywords: antibiotic resistance, Pakistan, carbapenem, multilocus sequence type, resistance

Introduction

Dissemination of carbapenem resistance in Gram-negative bacterial pathogens is a major problem worldwide.[1] Due to their broad-spectrum activity and resistance to degradation by extended-spectrum β-lactamases (ESBL), carbapenems have been used for treating severe infections caused by ESBL-positive bacteria for decades? However, the long use of this antibiotic has also decreased the effectiveness of this class of antibiotics.[2] It is estimated that ESBL prevalence in *Escherichia coli* is more than 80% in Pakistan (REF). This overuse of carbapenems has caused an increase in the number of carbapenem-resistant bacteria. Over the past 15 years, infections caused by carbapenem-resistant Enterobacteriaceae (CRE), such as *Klebsiella pneumoniae* and *E. coli* have increased.[1,2] Carbapenem-resistant *Acinetobacter baumannii* (CRAB) have become endemic and have been associated with high morbidity and mortality rates.[3] Similarly, the increase in carbapenem resistance in *Pseudomonas aeruginosa* has also become a serious challenge for clinicians.[4]

One of the most common mechanisms of carbapenem resistance is due to bacterial production of carbapenemases, carbapenem-hydrolyzing enzymes. Carbapenemase-encoding genes represent a global health threat because of their ability to be carried by mobile genetic elements.[1] Carbapenemases can be classified into various categories according to the Ambler classification scheme as A, B, and D.[5]

K. pneumoniae carbapenemases (KPC) are clinically the most common enzymes among the class A carbapenemases.[1,2] New Delhi metallo-β-lactamases (NDM), Verona integron-encoded metallo-β-lactamases (VIM), and active-on-imipenem (IMP) types belong to class B metallo-β-lactamases (MBLs),[1] whereas oxacillinases, including OXA-23, OXA-24, OXA-58, and OXA-48, are part of the class D enzymes.[1,2,5]

Previous studies have demonstrated prevalence of diverse carbapenemases in Gram-negative pathogens. In *A. baumannii*, class D oxacillinases are predominant, while class B carbapenemases were mostly found in *P. aeruginosa*.[5]

Besides producing carbapenem-degrading enzymes, there are various other mechanisms for the development of

[1]Faculté de Médecine et de Pharmacie, Aix Marseille Univ, IRD, APHM, MEPHI, IHU Méditerranée Infection, Marseille, France.
[2]Department of Microbiology, The University of Haripur, Haripur, Pakistan.
[3]Department of Pathology, Fauji Foundation Hospital, Rawalpindi, Pakistan.
AU3 ▶ [4]University of Lahore, Islamabad Campus, Islamabad, Pakistan.

resistance against carbapenems. For instance, it is believed that alteration in porin proteins plays a critical role. OprD is an outer membrane porin protein facilitating the excretion of carbapenems across the cell membrane. Alteration of the porin-coding *oprD* gene is also considered to be one of the main mechanisms conferring resistance to carbapenems in *P. aeruginosa*.[6]

In Pakistan, the high prevalence of ESBL-producing bacteria has led to the elevated use of carbapenems.[7,8] The most common carbapenemase-encoding gene was bla_{NDM}.[7] Some other studies also focused on the diversity of the different gene variants responsible for carbapenem resistance or studied the genetic relationship among the clinical isolates.[7–11]

The purpose of this study was to describe the diversity of carbapenem-resistant strains, their mechanisms of resistance to these antibiotics, as well as their clonal types in Gram-negative bacteria from clinical samples in Pakistan.

Materials and Methods

Bacterial isolates and species identification

Bacterial isolates were collected from patients attending Kidney Center (Al-Sayed Hospital) and Fauji Foundation Hospital, Rawalpindi, Pakistan, from May to December 2016. The samples were from urine, Foley catheter, pus, throat swab, endotracheal tube, bronchoalveolar lavage, wound swab, double-J stent, blood culture, central venous pressure tubes, and nasogastric tubes, and were cultured on blood agar and MacConkey's agar at 37°C for 24 hours.

Bacterial isolates were identified using matrix-assisted laser desorption and ionization time-of-flight mass spectrometry (Microflex; BrükerDaltonics, Bremen, Germany).[12]

Antimicrobial susceptibility testing

Antimicrobial susceptibility testing (AST) was initially performed by conventional Kirby Bauer disc diffusion to screen carbapenem-resistant isolates. Afterward, all carbapenem-resistant isolates were tested to determine minimum inhibitory concentrations (MIC). The MIC of imipenem and meropenem was determined by broth microdilution and by using E-test strips (bioMérieux, Marcy-l'Etoile, France). Colistin resistance was also determined by UMIC (Biocentric, Bandol, France) according to EUCAST recommendations. UMIC colistin is a ready-to-use broth microdilution kit for determining the MIC of colistin.

For Enterobacteriaceae strains, a panel of 16 antibiotics was tested, including amoxicillin, amoxicillin-clavulanic acid, cefepime, ceftriaxone, cefalothin, piperacillin-tazobactam, ertapenem, imipenem, meropenem, nitrofurantoin, fosfomycin, doxycycline, trimethoprim-sulfamethoxazole, amikacin, gentamicin, and ciprofloxacin. For non-Enterobacteriaceae isolates, other panels of 16 antibiotics consisting of ticarcillin, ticarcillin-clavulanic acid, cefepime, ceftazidime, piperacillin-tazobactam, imipenem, meropenem, nitrofurantoin, fosfomycin, doxycycline, trimethoprim-sulfamethoxazole, amikacin, tobramycin, gentamicin, ciprofloxacin, and rifampicin were used. Interpretations were made according to EUCAST breakpoints.[13]

DNA extraction

Bacterial DNA was extracted using the automatic robot EZ1 tool (Qiagen, Hilden, Germany), following the manufacturer's instructions. The extracted DNA was eluted in 200 μL elution buffer.

Determination of molecular mechanisms of antibiotic resistance

Real-time PCR was performed on all strains to screen for the presence of ESBL genes (bla_{CTX-M} group A [CTX-M-1 and CTX-M-9 groups], bla_{CTX-M} group B [CTX-M-2, CTX-M-8, and CTX-M-25 groups], bla_{TEM}, and bla_{SHV})[14] and carbapenem-hydrolyzing enzyme-encoding genes [bla_{NDM}, bla_{KPC}, bla_{OXA-48}, bla_{VIM}, bla_{OXA-23}, bla_{OXA-24}, and bla_{OXA-58}].[14,15]

Positive samples were further confirmed by standard PCR ◀AU5 and the amplicons were sequenced using the BigDye terminator chemistry on an ABI 3500 automated sequencer (Thermo Fisher Scientific, Waltham, MA). The sequences of the antibiotic resistance genes obtained were interpreted using the ARG-ANNOT database.[16]

Carbapenem resistance due to mutations in the *oprD* gene was investigated in *P. aeruginosa* isolates.[17] The sequences obtained were compared against PA01 *oprD* sequence AAG04347 using blastN and blastP.

Molecular typing by multilocus sequence typing

Molecular typing of *E. coli*, *Enterobacter cloacae*, *K. pneumoniae*, *P. aeruginosa*, and *Acinetobacter spp.* was performed to determine the genetic relationship among different clinical isolates using multilocus sequence typing (MLST).

Clonal types were determined for *E. coli*, *E. cloacae*, and *P. aeruginosa* in MLST Databases (http://mlst.warwick.ac.uk/mlst/dbs/Ecoli, https://pubmlst.org/ecloacae/, https://pubmlst.org/paeruginosa/) and according to the Pasteur schemes for *A. baumannii* and *K. pneumoniae* (bigsdb.web.pasteur.fr/).

Results

Bacteria strain identification

A total of 44 isolates displayed carbapenem resistance during the study period, including *K. pneumoniae* ($n = 13$), *E. coli* ($n = 9$), *A. baumannii* ($n = 10$), *P. aeruginosa* ($n = 6$), *E. cloacae* ($n = 4$), *Klebsiella oxytoca* ($n = 1$), and *Achromobacter xylosoxidans* ($n = 1$) (Table 1). ◀T1

AST of isolated strains

Antibiotic susceptibility testing results revealed high levels of resistance in most of the isolates to tested antibiotics (Table 1). Enterobacteriaceae isolates were resistant to all β-lactams, including imipenem (MIC ranging from 1.5 to >32 μg/mL). In addition, resistance to ciprofloxacin and aminoglycosides (amikacin and gentamicin) was also observed in all *K. pneumoniae* strains. Resistance to gentamicin and trimethoprim-sulfamethoxazole was observed in *E. cloacae* isolates. However, all *E. coli* strains were sensitive to fosfomycin, while 35.7% of *Klebsiella* spp. and 50% of *E. cloacae* were also sensitive (Table 1).

For non-Enterobacteriaceae isolates, all *A. baumannii* isolates were resistant to all tested β-lactams and also to imipenem, nitrofurantoin, trimethoprim-sulfamethoxazole, gentamicin, and ciprofloxacin. *A. xylosoxidans* was susceptible to ceftazidime and piperacillin-tazobactam, but

Table 1. Phenotypic and Genotypic Features of Enterobacteriaceae and Non-Enterobacteriaceae Clinical Isolates

Isolates	*Susceptible phenotype*	*Resistance phenotype*	*IMP MIC (μg/mL)*	*β-lactamase genes*	*Carbapenems resistance genes*	*Sequence type*
Enterobacteriaceae						
Escherichia coli 6, 12a, 24 ($n=3$)	FF, NF, DO, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, AK, GEN, CIP	>32	$bla_{CTX\text{-}M15}$ ($n=2$), $bla_{TEM\text{-}206}$ ($n=1$)	$bla_{NDM\text{-}1}$	167
E. coli 31a	FF, DO, SXT, CT	AMX, AMC, FEP, CRO, KF, TZP, NF, ERT, MEM, IMP, AK, GEN, CIP	>32	$bla_{TEM\text{-}206}$	$bla_{NDM\text{-}7}$	167
E. coli 26b, 27b ($n=2$)	FF, NF, GEN, DO, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, AK, CIP	4–16	$bla_{CTX\text{-}M15}$, $bla_{TEM\text{-}169}$	$bla_{OXA\text{-}48}$	617
E. coli 29	FF, NF, GEN, AK, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, DO, CIP	>32	$bla_{TEM\text{-}206}$	$bla_{NDM\text{-}5}$	405
E. coli 32	FF, NF, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, DO, AK, GEN, CIP	>32	$bla_{CTX\text{-}M15}$, $bla_{TEM\text{-}206}$	$bla_{OXA\text{-}181}$, $bla_{NDM\text{-}4}$	410
E. coli 41b	FF, NF, IMP, SXT, GEN, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, AK, DO, CIP	1.5	$bla_{CTX\text{-}M15}$	$bla_{OXA\text{-}181}$	410
Klebsiella pneumoniae 35B	FF, SXT, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, NF, DO, AK, GEN, CIP	>32	$bla_{CTX\text{-}M15}$, $bla_{SHV\text{-}28}$, $bla_{TEM\text{-}104}$	$bla_{NDM\text{-}1}$	15
K. pneumoniae 3a, 11a, 13b, 14b, 15b, 16a, 21b, 34, 40a ($n=9$)	CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, FF, NF, DO, AK, GEN, CIP	8–>32	$bla_{CTX\text{-}M15}$, $bla_{SHV\text{-}67}$, $bla_{TEM\text{-}206}$	$bla_{OXA\text{-}48}$	147
K. pneumoniae 33b, 43 ($n=2$)	FF, NF, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, DO, AK, GEN, CIP	4	$bla_{CTX\text{-}M15}$, $bla_{SHV\text{-}67}$, $bla_{TEM\text{-}206}$	$bla_{NDM\text{-}1}$	273
K. pneumoniae 36	FF, SXT, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, NF, DO, AK, GEN, CIP	6	$bla_{CTX\text{-}M15}$, $bla_{SHV\text{-}28}$, $bla_{TEM\text{-}206}$	$bla_{NDM\text{-}1}$	101
Klebsiella oxytoca 28	FF, NF, GEN, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, AK, DO, CIP	>32	$bla_{CTX\text{-}M15}$	$bla_{NDM\text{-}1}$	ND
Enterobacter cloacae 8	FF, NF, AK, DO, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, GEN, CIP	>32	$bla_{TEM\text{-}206}$	$bla_{NDM\text{-}1}$	177
E. cloacae 10	DO, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, FF, NF, AK, GEN, CIP	>32	$bla_{TEM\text{-}206}$	$bla_{NDM\text{-}1}$	177
E. cloacae 35A	FF, NF, AK, CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, DO, GEN, CIP	16	$bla_{TEM\text{-}104}$	$bla_{NDM\text{-}1}$	177
E. cloacae 25	CT	AMX, AMC, FEP, CRO, KF, TZP, ERT, MEM, IMP, SXT, FF, NF, DO, AK, GEN, CIP	>32	$bla_{CTX\text{-}M15}$, $bla_{TEM\text{-}198}$	$bla_{NDM\text{-}1}$	182

(*continued*)

Table 1. (Continued)

Isolates	*Susceptible phenotype*	*Resistance phenotype*	*IMP MIC (μg/mL)*	*β-lactamase genes*	*Carbapenems resistance genes*	*Sequence type*
Non-Enterobacteriaceae						
AU11 ▶ *Acinetobacter baumannii* 2, 12b, 4a, 38b ($n=4$)	DO, CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, FF, NF, SXT, TOB, AK, GEN, RA, CIP	>32	/	bla_{OXA-23}	2 ($n=2$), 1,106 ($n=1$), 1,209 ($n=1$)
A. baumannii 3	DO, FF, TOB, CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, NF, SXT, AK, GEN, RA, CIP	>32	/	bla_{OXA-23}	2
A. baumannii 7	DO, TOB, CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, FF, NF, SXT, AK, GEN, RA, CIP	>32	/	bla_{OXA-23}	2
A. baumannii 37a	CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, DO, FF, NF, SXT, TOB, AK, GEN, RA, CIP	>32	/	bla_{OXA-23}; bla_{OXA-72}	2
A. baumannii 5a, 9b ($n=2$)	DO, AK, CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, FF, NF, SXT, TOB, GEN, RA, CIP	>32	/	bla_{OXA-23}	1,106
A. baumannii 82	DO, TOB, CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, FF, NF, SXT, AK, GEN, RA, CIP	>32	/	bla_{OXA-23}, bla_{NDM-1}	1
Pseudomonas. aeruginosa 1, 19 ($n=2$)	CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, DO, FF, NF, SXT, TOB, AK, GEN, RA, CIP	>32	/	bla_{VIM-6}, lost oprD	815
P. aeruginosa 15	CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, DO, FF, NF, SXT, TOB, AK, GEN, RA, CIP	>32	/	bla_{VIM-11}, lost oprD	316
P. aeruginosa 13a, 15a, 18 ($n=3$)	CT	TIC, TIM, CAZ, FEP, TZP, MEM, IMP, DO, FF, NF, SXT, TOB, AK, GEN, RA, CIP	>32	/	stop oprD ($n=2$), lost oprD ($n=1$)	664
Achromobacter xylosoxidans 22	TIM, CAZ, TZP, SXT, CT	TIC, FEP, MEM, IMP, FF, NF, GEN, TOB, AK, DO, RA, CIP	8	/	/	ND

AU12 ▶ AK, amikacin; AMC, amoxicillin-clavulanic acid; AMX, amoxicillin; CAZ, ceftazidime; CIP, ciprofloxacin; CRO, ceftriaxone; CT, colistin; DO, doxycycline; ERT, ertapenem; FEP, cefepime; FF, fosfomycin; GEN, gentamicin; IMP, imipenem; KF, cefalothin; MEM, meropenem; MIC, minimum inhibitory concentrations; ND; NF, nitrofurantoin; RA, rifampicin; SXT, trimethoprim-sulfamethoxazole; TIC, ticarcillin; TIM, ticarcillin-clavulanic acid; TOB, tobramycin; TZP, piperacillin-tazobactam.

resistant to imipenem, fosfomycin, and aminoglycosides. Furthermore, all *P. aeruginosa* isolates were resistant to all antibiotics tested, except colistin. Colistin was found to be active against all isolates with MICs ranging from 0.125 to 1 μg/mL.

Molecular mechanisms of antibiotic resistance

The number of resistance genes in *E. coli* ($n = 9$) were as follows: ESBL genes $bla_{\text{TEM-206}} = 4$, $bla_{\text{TEM-169}} = 2$, and $bla_{\text{CTX-M-15}} = 7$, and carbapenem-hydrolyzing enzyme-encoding genes $bla_{\text{NDM-1}} = 3$, $bla_{\text{NDM-4}} = 1$, $bla_{\text{NDM-5}} = 1$, $bla_{\text{NDM-7}} = 1$, $bla_{\text{OXA-48}} = 2$, and $bla_{\text{OXA-181}} = 2$, whereas one strain #32 carried both $bla_{\text{OXA-181}}$ and $bla_{\text{NDM-4}}$ gene (Table 1).

Antibiotic resistance genes in the case of *Klebsiella* isolates ($n = 14$) were as follows: $bla_{\text{TEM-206}} = 12$, $bla_{\text{TEM-104}} = 1$, $bla_{\text{CTX-M-15}} = 14$, $bla_{\text{SHV-67}} = 10$, $bla_{\text{SHV-28}} = 2$, $bla_{\text{SHV-11}}$ $n = 1$, $bla_{\text{OXA-48}}$ $n = 9$, and $bla_{\text{NDM-1}} = 5$. $Bla_{\text{TEM-206}} = 2$, $bla_{\text{TEM-104}} = 1$, $bla_{\text{TEM-198}} = 1$, $bla_{\text{CTX-M-15}} = 1$, and $bla_{\text{NDM-1}} = 4$ genes were found in the four *E. cloacae* isolates (Table 1).

All imipenem-resistant *A. baumannii* isolates harbored $bla_{\text{OXA-23}}$gene, except isolates 37a and 80 that carried the$bla_{\text{OXA-72}}$ and $bla_{\text{NDM-1}}$ genes, respectively. In the case of six *P. aeruginosa* isolates, mutations leading to a stop codon on *oprD* gene were found in two isolates and the loss of *oprD* gene was observed in the other four isolates. In addition, carbapenem-hydrolyzing enzyme-encoding gene $bla_{\text{VIM-11}}$ was found in isolate 15 and $bla_{\text{VIM-6}}$ in isolates 1 and 19 (Table 1).

In summary, the frequency of antibiotic resistance genes in Enterobacteriaceae was as follows: $bla_{\text{NDM}} = 55.5\%$, $bla_{\text{OXA-48-like}} = 48.1\%$, $bla_{\text{TEM}} = 85.2\%$, $bla_{\text{CTX-M-15}} = 81.5\%$, and $bla_{\text{SHV}} = 48.1\%$. In *P. aeruginosa*, carbapenem resistance is due to the presence of bla_{VIM} gene (50%) and oprD modifications (100%). *A. baumannii* isolates harbored $bla_{\text{OXA-23}}$ gene (100%), $bla_{\text{OXA-72}}$ (10%), and bla_{NDM} genes (10%).

Molecular typing by MLST

AU6► From the MLST analysis, four STs were identified in *E. coli*, including ST167 ($n = 4$), ST617 ($n = 2$), ST410 ($n = 2$), and ST405 ($n = 1$). For *K. pneumoniae*, majority of STs were ST147 ($n = 9$) followed by ST273 ($n = 2$), ST101 ($n = 1$), and ST15 ($n = 1$). All *E. cloacae* belonged to two sequence types, that is, ST 177 ($n = 3$) and ST182 ($n = 1$) (Table 1).

Five STs were detected for *Acinetobacter* isolates (ST2 [$n = 5$], ST1106 [$n = 3$], ST207 [$n = 1$], ST1 [$n = 1$], and ST1209 [$n = 1$]). ST1209 is a new ST type assigned by the Pasteur MLST Database. In *P. aeruginosa*, ST664 ($n = 3$), ST815 ($n = 2$), and ST316 ($n = 1$) were found (Table 1).

Discussion

The results of this study show a high level of antibiotic resistance in clinical Gram-negative isolates against commonly used antibiotics in general and carbapenems in particular. In recent years, the rise in levels of antibiotic resistance, including increasing prevalence of ESBL-producing strains, has been reported from different areas of Pakistan. Therefore, carbapenems have become the drugs of choice to treat such cases. Increased and injudicious use of carbapenems in Pakistan have resulted in rise of carbapenem resistance,[8,18] which is also a cause for concern as the number of available therapeutic options is now limited. Carbapenem resistance is due to diverse mechanisms, including the production of carbapenemases in Gram-negative bacteria. Bacteria producing these enzymes can hydrolyze not only carbapenems but also other antibiotics such as penicillins, cephalosporins, and monobactams as well.[1,5] In addition, these hydrolyzing enzymes are spreading globally because of their prime location on plasmids.[1,2] It is therefore necessary to monitor resistance to detect emergence of different and new resistance mechanisms to prevent the dissemination of novel resistance genes to the rest of the world.[1,28] ◄AU7

As reported in previous studies, bla_{NDM} was also found to be the most prevalent carbapenemase-encoding gene in Enterobacteriaceae isolates.[10] The second most prevalent genes were $bla_{\text{OXA-48-like}}$ and $bla_{\text{OXA-181}}$.[9,10] $bla_{\text{OXA-181}}$ differs from $bla_{\text{OXA-48}}$ by only four amino acid sequences and has been reported from several countries, including the Indian subcontinent.[11] The concurrent presence of different combinations of $bla_{\text{OXA-48}}$ variants and bla_{NDM} genes has already been reported in America,[19] Singapore,[20] Saudi Arabia,[21] India,[22] and Pakistan.[9] It is detected primarily in *K. pneumoniae* strains, and in *E. coli* isolates. Interestingly, a diversity of carbapenemase-encoding genes has been observed, including the variants $bla_{\text{NDM-1}}$, $bla_{\text{NDM-4}}$, $bla_{\text{NDM-5}}$, $bla_{\text{NDM-7}}$, $bla_{\text{OXA-48}}$, and $bla_{\text{OXA-181}}$ in *E. coli* isolates. In the Indian subcontinent, the variants $bla_{\text{NDM-5}}$ and $bla_{\text{NDM-7}}$ have already been described.[23,24]

In Pakistan, only a few studies have described the clonal type of carbapenem-producing Gram-negative bacteria, making it difficult to monitor the emergence or spread of certain specific clones.[9,11,25] Previous studies have reported *K. pneumoniae* belonging to ST147,[11] ST11, ST14, ST15, ST101, and ST307.[9] In our study, multiple other clones like ST15, ST147, ST101, and ST273 were also detected. ST147, ST101, and ST15 are international clones responsible for the spread of ESBLs and also dissemination of carbapenemase-encoding genes.[26] ST273 is an emerging clone described in *K. pneumoniae* carbapenem-resistant strain.[27,28]

In *E. coli* isolates, the majority of isolates belonged to common ST complexes, such as ST10 complex (ST167 and ST617) followed by ST23 and ST405 complexes. In Pakistan, ST10 and ST405 have been described in NDM-producing *E. coli* and ST617 in ESBL-producing *E. coli* from migratory avian species.[25,29,30] Carbapenem-resistant *E. coli* strains belonging to ST167 and ST405 are present in the Indian subcontinent, notably in Nepal and India.[23,24] In Pakistan, the epidemiology of CRE seems similar to regional epidemiology in the Indian subcontinent.

A. baumannii is considered a leading cause of nosocomial infections throughout the world. Carbapenems are commonly used to treat these infections that result in the emergence and spread of carbapenem-resistant *A. baumannii* strains. In Pakistan, the frequency of CRAB is very high ranging from 62% to 100% in some hospitals.[18] This study confirms the predominant presence of CRAB carrying $bla_{\text{OXA-23}}$ and the emergence of strains harboring bla_{NDM} or $bla_{\text{OXA-72}}$.[10,31] MLST typing revealed that most carbapenem-resistant *A. baumannii* isolates belonged to the ST2 (CC2 complex) and some isolates were ST1 (CC1 complex), ST1106, and ST1209. *A. baumannii* infections worldwide belong to international complexes CC1, CC2, and CC3 and these CC are frequently linked to the production of OXA-23, OXA-24, or OXA-58 enzymes.[15,32] Alternatively, the rare ST1106 clone seems to be emerging in Pakistan where it has already been reported in clinical strains.[33] In addition, the detection of a new ST1209 clone in carbapenem-resistant *A. baumannii* isolates should be monitored.

Carbapenem-resistant *P. aeruginosa* are frequent in Pakistan. A previous study reported that 61.89% of *Pseudomonas* spp. were carbapenem-resistant *P. aeruginos*a isolates, including 78% of MBL producers in a Lahore hospital.[8] Genes encoding the VIM enzyme have also been detected in Pakistan, notably $bla_{\text{VIM-1}}$, $bla_{\text{VIM-2}}$, $bla_{\text{VIM-4}}$, and $bla_{\text{VIM-5}}$, but not $bla_{\text{VIM-6}}$ and $bla_{\text{VIM-11}}$, suggesting a change in the epidemiology of these genes.[10] These variants of bla_{VIM} are endemic in India, highlighting an epidemiology common to the Indian subcontinent.[34]

Carbapenem resistance due to the alteration of the porin-coding *oprD* gene has not been investigated in detail in Pakistan, although it is an important phenomenon. In this study, the presence of this mechanism, which can be associated with the presence of MBLs, has been observed with the loss of *oprD* or loss of expression by a stop codon. The population structure of *P. aeruginosa* is very diverse, but international high-risk clones, including ST664, are well known.[35] The other clones belonging to ST815 and ST316 observed in our study were less predominant, but were associated with carbapenem-resistant isolates found in the Gulf countries and in Portugal.[36,37]

In addition, the presence of multidrug-resistant opportunistic *A. xylosoxidans*, responsible for nosocomial infections,[38] should be monitored. To the best of our knowledge, we report here, for the first time, VIM variants (VIM-6 and VIM-11) in *P. aeruginosa* isolated from various clinical samples in Pakistan. Moreover, this is the only study in Pakistan describing the clonality of different Gram-negative bacteria by MLST, including *E. coli*, *K. pneumoniae*, *E. cloacae*, *Acinetobacter* spp., and *P. aeruginosa* isolates, thus allowing the description of the new ST1209 in *A. baumannii*.

Conclusion

This study highlights the diversity of Gram-negative bacteria resistant to carbapenems and the variety of resistance mechanisms that are associated with them in Pakistan.

The molecular structure of carbapenem resistance in Enterobacteriaceae and *P. aeruginosa* evolves with the emergence of different genes and gene variants encoding carbapenemases. Clone monitoring is essential to understand the evolution of a country's epidemiology and compare it with the regional and international epidemiology. In our case, although it is similar to the Indian subcontinent, some regional specificity in the clonal types was observed. The identification of high-risk clones with extensive multidrug resistance highlights the importance of a global surveillance of antimicrobial resistance.

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Disclosure Statement

No competing financial interests exist.

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References

1. Nordmann, P., and L. Poirel. 2014. The difficult-to-control spread of carbapenemase producers among Enterobacteriaceae worldwide. Clin. Microbiol. Infect. 20:821–830.
2. Lee, C.-R., J.H. Lee, K.S. Park, Y.B. Kim, B.C. Jeong, and S.H. Lee. 2016. Global dissemination of carbapenemase-producing *Klebsiella pneumoniae*: epidemiology, genetic context, treatment options, and detection methods. Front. Microbiol. 7:895.
3. Karampatakis, T., C. Antachopoulos, A. Tsakris, and E. Roilides. 2017. Molecular epidemiology of carbapenem-resistant *Acinetobacter baumannii* in Greece: an extended review (2000–2015). Future Microbiol. 12:801–815.
4. Pérez, A., E. Gato, J. Pérez-Llarena, F. Fernández-Cuenca, M.J. Gude, M. Oviaño, M.E. Pachón, J. Garnacho, V. González, Á. Pascual, J.M. Cisneros, and G. Bou. 2019. High incidence of MDR and XDR *Pseudomonas aeruginosa* isolates obtained from patients with ventilator-associated pneumonia in Greece, Italy and Spain as part of the MagicBullet clinical trial. J. Antimicrob. Chemother. 74:1244–1252.
5. Queenan, A.M., and K. Bush. 2007. Carbapenemases: the versatile β-lactamases. Clin. Microbiol. Rev. 20:440–458.
6. Telling, K., M. Laht, A. Brauer, M. Remm, V. Kisand, M. Maimets, T. Tenson, and I. Lutsar. 2018. Multidrug resistant *Pseudomonas aeruginosa* in Estonian hospitals. BMC Infect. Dis. 18:513.
7. Nahid, F., A.A. Khan, S. Rehman, and R. Zahra. 2013. Prevalence of metallo-β-lactamase NDM-1-producing multi-drug resistant bacteria at two Pakistani hospitals and implications for public health. J. Infect. Public Health 6:487–493.
8. Ain, N.U., A. Iftikhar, S.S. Bukhari, S. Abrar, S. Hussain, M.H. Haider, F. Rasheed, and S. Riaz. 2018. High frequency and molecular epidemiology of metallo-β-lactamase-producing gram-negative bacilli in a tertiary care hospital in Lahore, Pakistan. Antimicrob. Resist. Infect. Control 7:128.
9. Lomonaco, S., M.A. Crawford, C. Lascols, R.E. Timme, K. Anderson, D.R. Hodge, D.J. Fisher, S.P. Pillai, S.A. Morse, E. Khan, M.A. Hughes, M.W. Allard, and S.K. Sharma. 2018. Resistome of carbapenem- and colistin-resistant *Klebsiella pneumoniae* clinical isolates. PLoS One 13:e0198526.
10. Braun, S.D., B. Jamil, M.A. Syed, S.A. Abbasi, D. Weiß, P. Slickers, S. Monecke, I. Engelmann, and R. Ehricht. 2018. Prevalence of carbapenemase-producing organisms at the Kidney Center of Rawalpindi (Pakistan) and evaluation of an advanced molecular microarray-based carbapenemase assay. Future Microbiol. 13:1225–1246.
11. Nahid, F., R. Zahra, and L. Sandegren. 2017. A blaOXA-181-harbouring multi-resistant ST147 Klebsiella pneumo-

niae isolate from Pakistan that represent an intermediate stage towards pan-drug resistance. PLoS One 12:e0189438.
12. Seng, P., M. Drancourt, F. Gouriet, B. La Scola, P.-E. Fournier, J.M. Rolain, and D. Raoult. 2009. Ongoing revolution in bacteriology: routine identification of bacteria by matrix-assisted laser desorption ionization time-of-flight mass spectrometry. Clin. Infect. Dis. 49: 543–551.
13. EUCAST. 2017. European Committee on Antimicrobial Susceptibility Testing Breakpoint tables for interpretation of MICs and zone diameters. 7.1. Available at www.eucast .org/fileadmin/src/media/PDFs/EUCAST_files/Breakpoint_tables/v_50_Breakpoint_Table_01.pdf
14. Yousfi, H., L. Hadjadj, I. Dandachi, R. Lalaoui, A. Merah, K. Amoura, A. Dahi, M. Dekhil, N. Messalhi, S.M. Diene, S. Baron, and J.-M. Rolain. 2019. Colistin- and carbapenem-resistant *Klebsiella pneumoniae* clinical_isolates: algeria. Microb. Drug Resist. 25:258–263.
15. Nawfal Dagher, T., C. Al-Bayssari, S. Chabou, N. Antar, S.M. Diene, E. Azar, and J.-M. Rolain. 2019. Investigation of multidrug-resistant ST2 *Acinetobacter baumannii* isolated from Saint George hospital in Lebanon. BMC Microbiol. 19:29.
16. Gupta, S.K., B.R. Padmanabhan, S.M. Diene, R. Lopez-Rojas, M. Kempf, L. Landraud, and J.-M. Rolain. 2014. ARG-ANNOT, a new bioinformatic tool to discover antibiotic resistance genes in bacterial genomes. Antimicrob. Agents Chemother. 58:212–220.
17. Nawfal Dagher, T., C. Al-Bayssari, S.M. Diene, E. Azar, and J.-M. Rolain. 2019. Emergence of plasmid-encoded VIM-2-producing *Pseudomonas aeruginosa* isolated from clinical samples in Lebanon. New Microbes New Infect. 29:100521.
18. Asif, M., I.A. Alvi, and S.U. Rehman. 2018. Insight into *Acinetobacter baumannii*: pathogenesis, global resistance, mechanisms of resistance, treatment options, and alternative modalities. Infect. Drug Resist. 11:1249–1260.
19. Rojas, L.J., A.M. Hujer, S.D. Rudin, M.S. Wright, T.N. Domitrovic, S.H. Marshall, K.M. Hujer, S.S. Richter, E. Cober, F. Perez, M.D. Adams, D. van Duin, and R.A. Bonomo. 2017. NDM-5 and OXA-181 beta-lactamases, a significant threat continues to spread in the Americas. Antimicrob. Agents Chemother. 61. DOI: 10.1128/aac.00454-17
20. Balm, M.N.D., M.-V. La, P. Krishnan, R. Jureen, R.T.P. Lin, and J.W.P. Teo. 2013. Emergence of *Klebsiella pneumoniae* co-producing NDM-type and OXA-181 carbapenemases. Clin. Microbiol. Infect. 19:E421–E423.
21. Al-Zahrani, I.A., and B.A. Alsiri. 2018. The emergence of carbapenem-resistant Klebsiella pneumoniae isolates producing OXA-48 and NDM in the Southern (Asir) province, Saudi Arabia. Saudi Med. J. 39:23–30.
22. Khajuria, A. 2014. Emergence of *Escherichia coli*, co-producing NDM-1 and OXA-48 carbapenemases, in urinary isolates, at a tertiary care centre at Central India. J. Clin. Diagn. Res. 8:DC01–DC04.
23. Ranjan, A., S. Shaik, A. Mondal, N. Nandanwar, A. Hussain, T. Semmler, N. Kumar, S.K. Tiwari, S. Jadhav, L.H. Wieler, and N. Ahmed. 2016. Molecular epidemiology and genome dynamics of New Delhi metallo-β-lactamase-producing extraintestinal pathogenic *Escherichia coli* strains from India. Antimicrob. Agents Chemother. 60:6795–6805.
24. Shrestha, B., T. Tada, K. Shimada, S. Shrestha, H. Ohara, B.M. Pokhrel, J.B. Sherchand, and T. Kirikae. 2017. Emergence of various NDM-type-metallo-β-lactamase-producing *Escherichia coli* clinical isolates in Nepal. Antimicrob. Agents Chemother. 61:e01425-17.
25. Sartor, A.L., M.W. Raza, S.A. Abbasi, K.M. Day, J.D. Perry, D.L. Paterson, and H.E. Sidjabat. 2014. Molecular epidemiology of NDM-1-producing Enterobacteriaceae and *Acinetobacter baumannii* isolates from Pakistan. Antimicrob. Agents Chemother. 58:5589–5593.
26. Liapis, E., A. Pantel, J. Robert, M.-H. Nicolas-Chanoine, L. Cavalié, N. van der Mee-Marquet, C. de Champs, N. Aissa, C. Eloy, V. Blanc, C. Guyeux, D. Hocquet, J.-P. Lavigne, X. Bertrand, and On behalf of ONERBA. 2014. Molecular epidemiology of OXA-48-producing *Klebsiella pneumoniae* in France. Clin. Microbiol. Infect. 20: O1121– O1123.
27. Liu, L., Y. Feng, H. Long, A. McNally, and Z. Zong. 2018. Sequence type 273 Carbapenem-resistant *Klebsiella pneumoniae* carrying blaNDM-1 and blaIMP-4. Antimicrob. Agents Chemother. 62:e00160-18.
28. Qamar, M.U., T.R. Walsh, M.A. Toleman, J.M. Tyrrell, S. Saleem, A. Aboklaish, and S. Jahan. 2019. Dissemination of genetically diverse NDM-1, -5, -7 producing-Gram-negative pathogens isolated from pediatric patients in Pakistan. Future Microbiol. 14:691–704.
29. Mohsin, M., S. Raza, K. Schaufler, N. Roschanski, F. Sarwar, T. Semmler, P. Schierack, and S. Guenther. 2017. High prevalence of CTX-M-15-type ESBL-producing *E. coli* from migratory avian species in Pakistan. Front. Microbiol. 8:2476.
30. Qamar, M.U., T.R. Walsh, M.A. Toleman, S. Saleem, and S. Jahan. 2018. First identification of clinical isolate of a Novel ''NDM-4'' producing *Escherichia coli* ST405 from urine sample in Pakistan. Braz. J. Microbiol. 49: 949–950.
31. Khurshid, M., M.H. Rasool, U.A. Ashfaq, B. Aslam, and M. Waseem. 2017. Emergence of ISAba1 harboring carbapenem-resistant *Acinetobacter baumannii* isolates in Pakistan. Future Microbiol. 12:1261–1269.
32. Karah, N., A. Sundsfjord, K. Towner, and Ø. Samuelsen. 2012. Insights into the global molecular epidemiology of carbapenem non-susceptible clones of *Acinetobacter baumannii*. Drug Resist. Updat. 15:237–247.
33. Ahmad, I., N. Karah, A. Nadeem, S.N. Wai, and B.E. Uhlin. 2019. Analysis of colony phase variation switch in *Acinetobacter baumannii* clinical isolates. PLoS One 14: e0210082.
34. Castanheira, M., J.M. Bell, J.D. Turnidge, D. Mathai, and R.N. Jones. 2009. Carbapenem resistance among *Pseudomonas aeruginosa* strains from India: evidence for nationwide endemicity of multiple metallo-beta-lactamase clones (VIM-2, -5, -6, and -11 and the newly characterized VIM-18). Antimicrob. Agents Chemother. 53:1225–1227.
35. Pragasam, A., B. Veeraraghavan, S. Anandan, V. Narasiman, S. Sistla, A. Kapil, P. Mathur, P. Ray, C. Wattal, S. Bhattacharya, V. Deotale, K. Subramani, J. Peter, T. Hariharan, I. Ramya, S. Iniyan, K. Walia, and V. Ohri. 2018. Dominance of international high-risk clones in carbapenemase-producing *Pseudomonas aeruginosa*: multicentric molecular epidemiology report from India. Indian J. Med. Microbiol. 36:344.
36. Botelho, J., F. Grosso, S. Quinteira, M. Brilhante, H. Ramos, and L. Peixe. 2018. Two decades of blaVIM-2-producing *Pseudomonas aeruginosa* dissemination: an

interplay between mobile genetic elements and successful clones. J. Antimicrob. Chemother. 73:873–882.

37. Zowawi, H.M., M.W. Syrmis, T.J. Kidd, H.H. Balkhy, T.R. Walsh, S.M. Al Johani, R.Y. Al Jindan, M. Alfaresi, E. Ibrahim, A. Al-Jardani, J. Al Salman, A.A. Dashti, H.E. Sidjabat, O. Baz, E. Trembizki, D.M. Whiley, and D.L. Paterson. 2018. Identification of carbapenem-resistant *Pseudomonas aeruginosa* in selected hospitals of the Gulf Cooperation Council States: dominance of high-risk clones in the region. J. Med. Microbiol. 67:846–853.
38. Abbott, I., and A. Peleg. 2015. Stenotrophomonas, Achromobacter, and Nonmelioid Burkholderia Species: antimicrobial resistance and therapeutic strategies. Semin. Respir. Crit. Care Med. 36:099-110.
39. Sahu, M.K., U. Balasubramaniam, B. C, S.P. Singh, and S. Talwar. 2019. Elizabethkingia meningoseptica: an emerging nosocomial pathogen causing septicemia in critically ill patients. Indian J. Crit. Care Med. 23:104–105. ◀AU9

Address correspondence to:
Bushra Jamil ◀AU10
University of Lahore
Islamabad Campus
Islamabad 46000
Pakistan

E-mail: bushrajamil2000@gmail.com

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